

QUITE UNIVERSAL CIRCUIT SIMULATOR: A Tool That Justifies Its Name

Quite Universal Circuit Simulator (QUCS) software makes electronic circuit simulation easy. That means you can digitally design your circuit and evaluate its functionality before creating the real circuit. It is a free software licensed under the GNU General Public License (GPL) that saw a stable release recently with many upgrades and improvements. It is easy to use and provides clear visibility of everything in your circuit.

What QUCS offers

QUCS brings an interactive graphical user interface (GUI) that helps to perform all the changes in a project on the workspace. Users can set up schematics for simulations, perform modifications or write codes on the interface as required. The software interface is quite familiar: The centre is occupied by workspace, where the designed circuit can be visualised using a netlist. Essential menus and tools on the top allow quick access to different functionalities. Main menus include File, Edit, Insert, Project, Simulation, View and Help. Below the menus are some quick access options for different circuit components, and view or file options. On the left of the workspace, a panel consists of information like available schematics, datasets and data displays. Each category can be expanded to access available elements.

QUCS involves a lot of command-line programming for setting directions of the operations to be performed, mainly for the analogue

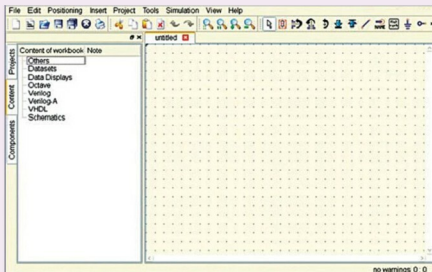


Fig. 1: Empty project while starting up QUCS

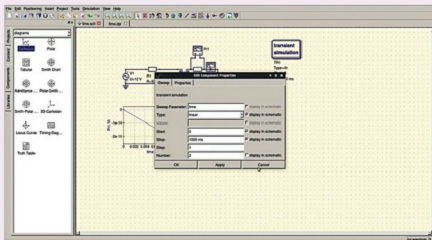


Fig. 2: The chart library and different component properties in QUCS

simulator. The tool feeds on pre-built netlists, checks errors and proceeds for simulation. Data generated from simulation is stored as a dataset.

QUCS can run small-signal as well as large-signal circuits. It also supports SPICE sub-circuits. Even

languages like VHDL or Verilog are supported for pure digital circuit simulation.

Components library manager in QUCS preserves all the digital replica of real-life electronic components like diodes, transistors, op-amps and